NETWORKING HOME LAB PORTFOLIO

Hands-on configuration • Verification • Troubleshooting

Project 1	MikroTik Routing and VLAN Lab
Project 2	Ubuntu Server and WireGuard VPN
Project 3	UniFi Wireless Network Integration
Project 4	Cisco Packet Tracer Networking Labs

Prepared as a practical learning portfolio for entry-level Network Technician, IT Support, and Junior Network Engineer opportunities.

This document records practical lab work and the troubleshooting methods used. It distinguishes confirmed observations from incomplete work.

Project 3 — UniFi Wireless Network Integration

Wireless networking | Ubiquiti UniFi AP-AC-Pro and MikroTik hAP ac lite

1. Project Overview

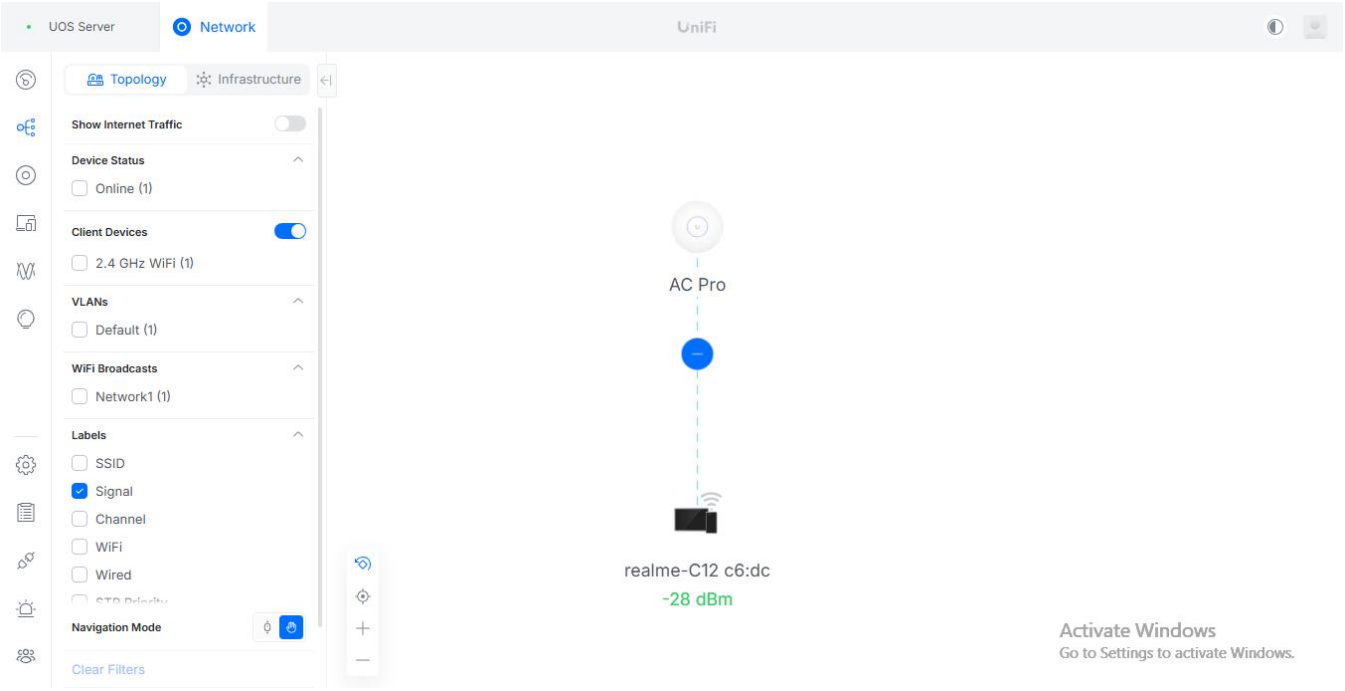
Connected and tested a UniFi AC Pro access point on a MikroTik hAP ac lite network. Verified that the AP obtained a LAN address, responded to ping, and broadcast Wi-Fi. Also investigated UniFi management/adoption and inform communication issues.

2. Objectives

- Connect a UniFi AP to the existing router LAN and confirm link/IP connectivity.
- Access the AP for diagnostics using SSH and inspect its device information.
- Check wireless broadcasting and management/controller status.
- Understand that assigning VLANs to SSIDs requires matching AP settings and switch/router trunk configuration.

3. Network Topology

Internet → MikroTik hAP ac lite (wireless uplink/LAN) → Ethernet → UniFi UAP-AC-Pro. Windows workstation is used for ping, browser management, and SSH diagnostics. Additional SSID-to-VLAN mapping is a planned extension and must not be assumed operational.



4. Configuration and Commands

The following are examples/reference commands. Adapt interface names, IPs, VLAN IDs, and device versions to the configuration you actually tested.

Windows reachability checks

```
ping <AP_IP>
arp -a
Test-NetConnection <AP_IP> -Port 22
```

UniFi AP information command (from AP SSH)

```
info
```

Controller inform URL example from this lab

```
http://192.168.88.254:8080/inform
Use only if this is still the correct controller address and port in your current network.
```

5. Verification and Testing

- Check the AP link and DHCP/IP assignment in the router or client list.

```
[admin@MikroTik] > /interface print
Flags: X - DISABLED; R - RUNNING; S - SLAVE
Columns: NAME, TYPE, ACTUAL-MTU, L2MTU, MAX-L2MTU, MAC-ADDRESS
#     NAME         TYPE      ACTUAL-MTU  L2MTU  MAX-L2MTU  MAC-ADDRESS
0  RS ether1       ether           1500   1598       2028  C4:AD:34:05:B5:1E
1  RS ether2       ether           1500   1598       2028  C4:AD:34:05:B5:1F
2   S ether3       ether           1500   1598       2028  C4:AD:34:05:B5:20
3   S ether4       ether           1500   1598       2028  C4:AD:34:05:B5:21
4   S ether5       ether           1500   1598       2028  C4:AD:34:05:B5:22
5 X   wlan1        wlan            1500   1600       2290  C4:AD:34:05:B5:24
6  R  wlan2        wlan            1500   1600       2290  C4:AD:34:05:B5:23
7  R  bridge-LAN   bridge          1500   1598             C4:AD:34:05:B5:20
;;; defconf
8  R  bridgeLocal  bridge          1500  65535             C4:AD:34:05:B5:1E
9  R  lo           loopback       65536                   00:00:00:00:00:00
```

```
[admin@MikroTik] > /ip dhcp-server lease print detail
Flags: D - DYNAMIC
0  D address=192.168.110.254 mac-address=CC:2D:E0:BB:AB:79
     client-id="1:cc:2d:e0:bb:ab:79" address-lists="" server=LAN_DHCP
     dhcp-option="" agent-circuit-id="" agent-remote-id="" status=bound
     expires-after=5m31s last-seen=1h40m3s active-address=192.168.110.254
     active-mac-address=CC:2D:E0:BB:AB:79 active-client-id="1:cc:2d:e0:bb:ab:79"
     active-server=LAN_DHCP active-agent-circuit-id="" active-agent-remote-id=""
     host-name="MikroTik"

1  D address=192.168.110.253 mac-address=14:DA:E9:66:CC:A2
     client-id="1:14:da:e9:66:cc:a2" address-lists="" server=LAN_DHCP
     dhcp-option="" agent-circuit-id="" agent-remote-id="" status=bound
     expires-after=7m10s last-seen=2m50s age=2m51s
     active-address=192.168.110.253 active-mac-address=14:DA:E9:66:CC:A2
     active-client-id="1:14:da:e9:66:cc:a2" active-server=LAN_DHCP
     active-agent-circuit-id="" active-agent-remote-id=""
     host-name="DESKTOP-C7HCUCL" class-id="MSFT 5.0"

2  D address=192.168.110.252 mac-address=80:2A:A8:C6:AD:EF
     client-id="1:80:2a:a8:c6:ad:ef" address-lists="" server=LAN_DHCP
     dhcp-option="" agent-circuit-id="" agent-remote-id="" status=bound
     expires-after=9m23s last-seen=37s age=38s active-address=192.168.110.252
     active-mac-address=80:2A:A8:C6:AD:EF active-client-id="1:80:2a:a8:c6:ad:ef"
     active-server=LAN_DHCP active-agent-circuit-id="" active-agent-remote-id=""
     class-id="ubnt"
```

```
[admin@MikroTik] > arp print detail
bad command name arp (line 1 column 1)
[admin@MikroTik] > ip arp print detail
Flags: D - DYNAMIC; C - COMPLETE
0   C address=192.168.88.250 mac-address=94:C6:91:11:DB:7A interface=bridge-LAN published=no status="permanent" vrf=main

1  DC address=192.168.110.253 mac-address=14:DA:E9:66:CC:A2 interface=bridge-LAN published=no status="reachable" vrf=main

2  DC address=192.168.0.112 mac-address=40:AE:30:B5:C1:2E interface=wlan2 published=no status="stale" vrf=main

3  DC address=192.168.110.252 mac-address=80:2A:A8:C6:AD:EF interface=bridge-LAN published=no status="reachable" vrf=main

4  D  address=192.168.0.111 interface=wlan2 published=no status="failed" vrf=main

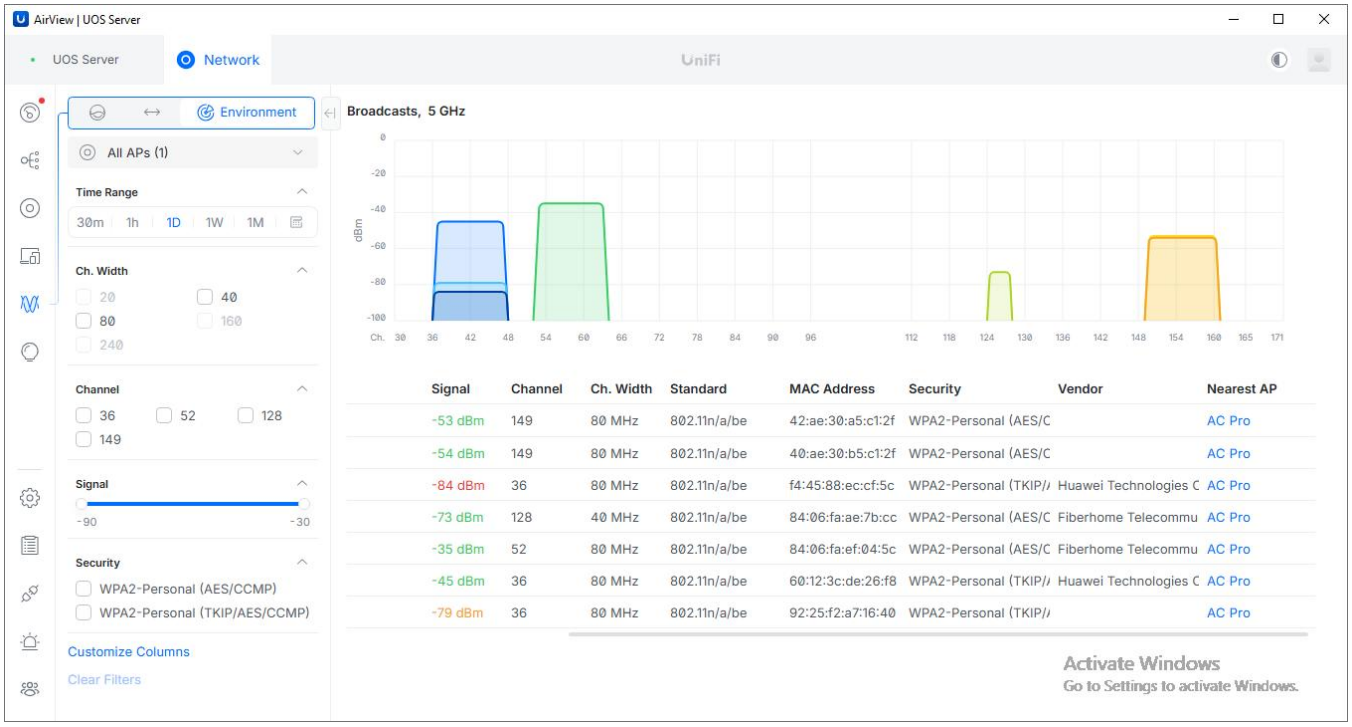
5  DC address=192.168.0.1 mac-address=3C:6A:D2:12:76:7B interface=wlan2 published=no status="reachable" vrf=main
[admin@MikroTik] >
```

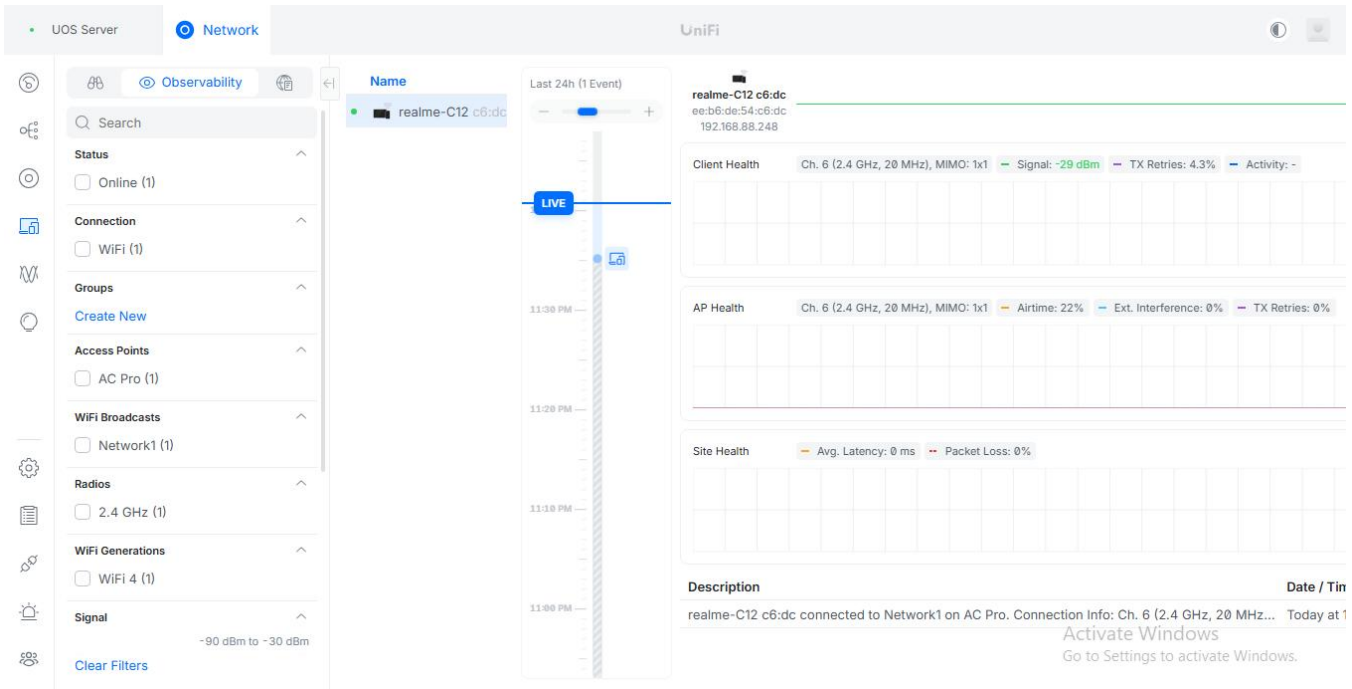
```
[admin@MikroTik] > /interface ethernet monitor ether2 once
                      name: ether2
                    status: link-ok
          auto-negotiation: done
                      rate: 100Mbps
               full-duplex: yes
            tx-flow-control: no
            rx-flow-control: no
                 supported: 10M-baseT-half
                            10M-baseT-full
                            100M-baseT-half
                            100M-baseT-full
               advertising: 10M-baseT-half
                            10M-baseT-full
                            100M-baseT-half
                            100M-baseT-full
  link-partner-advertising: 10M-baseT-half
                            10M-baseT-full
                            100M-baseT-half
                            100M-baseT-full
[admin@MikroTik] >
```

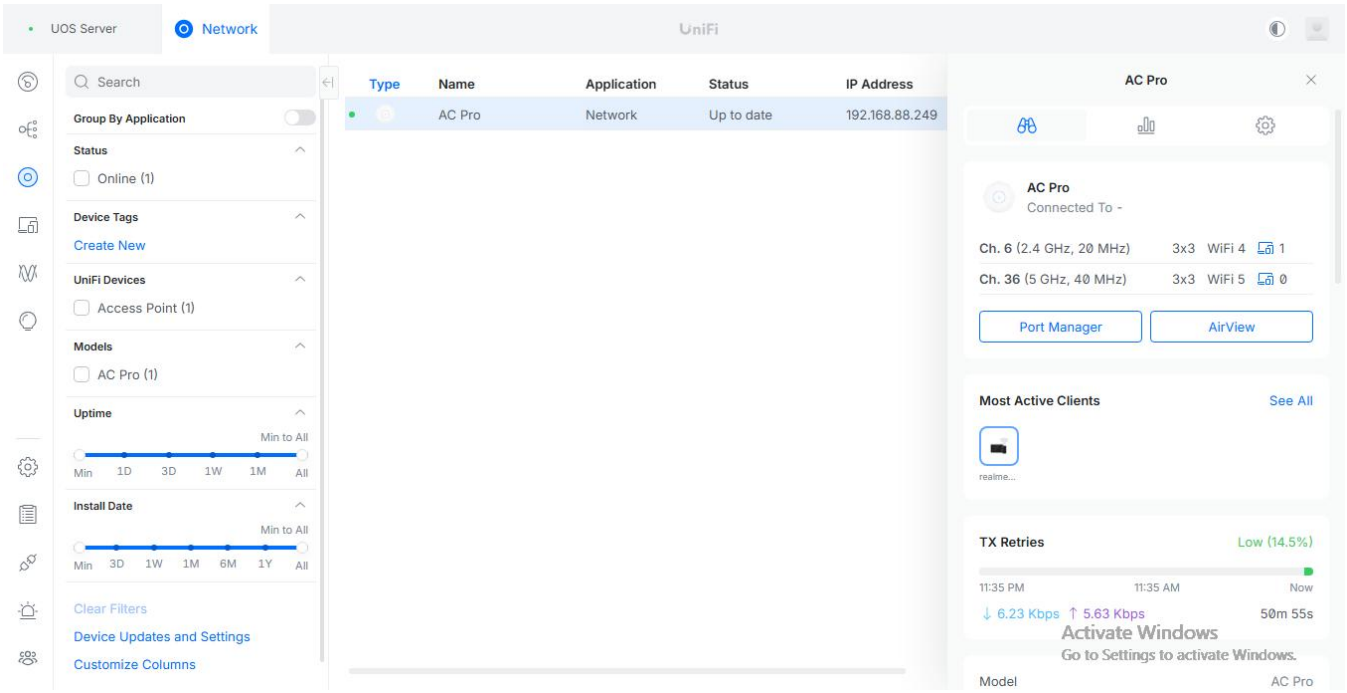
- Ping the AP IP from the Windows workstation and verify the ARP entry.

```
UAP-AC-Pro-Gen2-BZ.6.8.2# ping 192.168.110.252
PING 192.168.110.252 (192.168.110.252): 56 data bytes
64 bytes from 192.168.110.252: seq=0 ttl=64 time=0.251 ms
64 bytes from 192.168.110.252: seq=1 ttl=64 time=0.255 ms
64 bytes from 192.168.110.252: seq=2 ttl=64 time=0.239 ms
64 bytes from 192.168.110.252: seq=3 ttl=64 time=0.239 ms
64 bytes from 192.168.110.252: seq=4 ttl=64 time=0.241 ms
^C
--- 192.168.110.252 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0.239/0.245/0.255 ms
UAP-AC-Pro-Gen2-BZ.6.8.2# arp -a
? (192.168.110.253) at 14:da:e9:66:cc:a2 [ether]  on br0
? (192.168.110.1) at c4:ad:34:05:b5:20 [ether]  on br0
UAP-AC-Pro-Gen2-BZ.6.8.2#
```

- Confirm the AP management page/status is reachable and Wi-Fi is broadcasting.


AirView | UOS Server
UOS Server
Network
UniFi
Environment
All APs (1)
Time Range
30m 1h 1D 1W 1M
Ch. Width
20 40
80 160
240
Channel
36 52 128
149
Signal
-90 -30
Security
WPA2-Personal (AES/CCMP)
WPA2-Personal (TKIP/AES/CCMP)
Customize Columns
Clear Filters
Broadcasts, 5 GHz
dBm
Ch. 30 36 42 48 54 60 66 72 78 84 90 96 112 118 124 130 136 142 148 154 160 165 171
Signal Channel Ch. Width Standard MAC Address Security Vendor Nearest AP
-53 dBm 149 80 MHz 802.11n/a/be 42:ae:30:a5:c1:2f WPA2-Personal (AES/C AC Pro
-54 dBm 149 80 MHz 802.11n/a/be 40:ae:30:b5:c1:2f WPA2-Personal (AES/C AC Pro
-84 dBm 36 80 MHz 802.11n/a/be f4:45:88:ec:cf:5c WPA2-Personal (TKIP/ Huawei Technologies C AC Pro
-73 dBm 128 40 MHz 802.11n/a/be 84:06:fa:ae:7b:cc WPA2-Personal (AES/C Fiberhome Telecommu AC Pro
-35 dBm 52 80 MHz 802.11n/a/be 84:06:fa:ef:04:5c WPA2-Personal (AES/C Fiberhome Telecommu AC Pro
-45 dBm 36 80 MHz 802.11n/a/be 60:12:3c:de:26:f8 WPA2-Personal (TKIP/ Huawei Technologies C AC Pro
-79 dBm 36 80 MHz 802.11n/a/be 92:25:f2:a7:16:40 WPA2-Personal (TKIP/ AC Pro
Activate Windows
Go to Settings to activate Windows.


UOS Server
Network
UniFi
Observability
Search
Status
Online (1)
Connection
WiFi (1)
Groups
Create New
Access Points
AC Pro (1)
WiFi Broadcasts
Network1 (1)
Radios
2.4 GHz (1)
WiFi Generations
WiFi 4 (1)
Signal
-90 dBm to -30 dBm
Clear Filters
Name
realme-C12 c6:dc
Last 24h (1 Event)
LIVE
11:30 PM
11:20 PM
11:10 PM
11:00 PM
realme-C12 c6:dc
ee:b6:de:54:c6:dc
192.168.88.248
Client Health
Ch. 6 (2.4 GHz, 20 MHz), MIMO: 1x1
Signal: -29 dBm
TX Retries: 4.3%
Activity: -
AP Health
Ch. 6 (2.4 GHz, 20 MHz), MIMO: 1x1
Airtime: 22%
Ext. Interference: 0%
TX Retries: 0%
Site Health
Avg. Latency: 0 ms
Packet Loss: 0%
Description
Date / Tin
realme-C12 c6:dc connected to Network1 on AC Pro. Connection Info: Ch. 6 (2.4 GHz, 20 MHz... Today at
Activate Windows
Go to Settings to activate Windows.



- Check controller adoption state and inform URL; verify the controller is listening on the required ports.

```
UAP-AC-Pro-Gen2-BZ.6.8.2# Get-NetNeighbor -IPAddress 192.168.110.252
-ash: Get-NetNeighbor: not found
UAP-AC-Pro-Gen2-BZ.6.8.2# info

Model:       UAP-AC-Pro-Gen2
Version:     6.8.2.15592
MAC Address: 80:2a:a8:c6:ad:ef
IP Address:  192.168.110.252
Hostname:    UAP-AC-Pro-Gen2
Uptime:      1363 seconds
NTP:         Synchronized

Status:      Connected (http://192.168.110.253:8080/inform)
UAP-AC-Pro-Gen2-BZ.6.8.2# mca-ctrl -t dump-cfg
```

```
PS C:\Users\TAKLE> Get-NetTCPConnection -State Listen |
>> Where-Object { $_.LocalPort -in 8080,8443,8880,8843,3478 } |
>> Select-Object LocalAddress,LocalPort,OwningProcess

LocalAddress LocalPort OwningProcess
------------ --------- -------------
::1               8880         14440
::1               8080         14440
0.0.0.0           8880          4148
0.0.0.0           8080          4148
0.0.0.0           3478          4148
```

```
PS C:\Users\TAKLE> netstat -ano | findstr ":8080"
  TCP    0.0.0.0:8080           0.0.0.0:0              LISTENING       4148
  TCP    192.168.32.1:52909     192.168.47.48:8080     TIME_WAIT       0
  TCP    192.168.32.1:52910     192.168.47.48:8080     TIME_WAIT       0
  TCP    192.168.32.1:52911     192.168.47.48:8080     TIME_WAIT       0
  TCP    192.168.32.1:52913     192.168.47.48:8080     TIME_WAIT       0
  TCP    192.168.32.1:52914     192.168.47.48:8080     TIME_WAIT       0
  TCP    192.168.32.1:52917     192.168.47.48:8080     TIME_WAIT       0
  TCP    192.168.32.1:52918     192.168.47.48:8080     TIME_WAIT       0
  TCP    192.168.32.1:52919     192.168.47.48:8080     TIME_WAIT       0
  TCP    192.168.32.1:52922     192.168.47.48:8080     TIME_WAIT       0
  TCP    [::1]:8080             [::]:0                 LISTENING       14440
PS C:\Users\TAKLE>
```

6. Troubleshooting Experience

- Verified that a UniFi AP at a 192.168.110.x address responded to ping and was reachable from the Windows workstation.

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\TAKLE>ping 192.168.110.252

Pinging 192.168.110.252 with 32 bytes of data:
Reply from 192.168.110.252: bytes=32 time<1ms TTL=64
Reply from 192.168.110.252: bytes=32 time=1ms TTL=64
Reply from 192.168.110.252: bytes=32 time<1ms TTL=64
Reply from 192.168.110.252: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.110.252:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\TAKLE>arp -a

Interface: 192.168.110.253 --- 0x7
  Internet Address      Physical Address      Type
  192.168.110.1         c4-ad-34-05-b5-20     dynamic
  192.168.110.252       80-2a-a8-c6-ad-ef     dynamic
  192.168.110.255       ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  233.89.188.1          01-00-5e-59-bc-01     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

```
[admin@MikroTik] > /ip arp print detail
Flags: D - DYNAMIC; C - COMPLETE
0   C address=192.168.88.250 mac-address=94:C6:91:11:DB:7A interface=bridge-LAN published=no status="permanent" vrf=main

1  DC address=192.168.110.253 mac-address=14:DA:E9:66:CC:A2 interface=bridge-LAN published=no status="reachable" vrf=main

2  DC address=192.168.0.112 mac-address=40:AE:30:B5:C1:2E interface=wlan2 published=no status="stale" vrf=main

3  DC address=192.168.110.252 mac-address=80:2A:A8:C6:AD:EF interface=bridge-LAN published=no status="reachable" vrf=main

4  D  address=192.168.0.111 interface=wlan2 published=no status="failed" vrf=main

5  DC address=192.168.0.1 mac-address=3C:6A:D2:12:76:7B interface=wlan2 published=no status="reachable" vrf=main
[admin@MikroTik] > /ip dhcp-server lease print detail
Flags: D - DYNAMIC
0  D address=192.168.110.253 mac-address=14:DA:E9:66:CC:A2 client-id="1:14:da:e9:66:cc:a2" address-lists="" server=LAN_DHCP dhcp-option="" agent-circuit-id="" agent-remote-id=""
     status=bound expires-after=8m41s last-seen=1m19s age=41m20s active-address=192.168.110.253 active-mac-address=14:DA:E9:66:CC:A2 active-client-id="1:14:da:e9:66:cc:a2"
     active-server=LAN_DHCP active-agent-circuit-id="" active-agent-remote-id="" host-name="DESKTOP-C7HCUCL" class-id="MSFT 5.0"

1  D address=192.168.110.252 mac-address=80:2A:A8:C6:AD:EF client-id="1:80:2a:a8:c6:ad:ef" address-lists="" server=LAN_DHCP dhcp-option="" agent-circuit-id="" agent-remote-id=""
     status=bound expires-after=8m29s last-seen=1m31s age=11m32s active-address=192.168.110.252 active-mac-address=80:2A:A8:C6:AD:EF active-client-id="1:80:2a:a8:c6:ad:ef"
     active-server=LAN_DHCP active-agent-circuit-id="" active-agent-remote-id="" class-id="ubnt"
[admin@MikroTik] > /interface bridge host print detail
Flags: D - DYNAMIC; L - LOCAL; E - EXTERNAL
0 DL  mac-address=C4:AD:34:05:B5:1E interface=bridgeLocal bridge=bridgeLocal on-interface=bridgeLocal

1 D E mac-address=14:DA:E9:66:CC:A2 interface=ether2 bridge=bridge-LAN on-interface=ether2

2 D E mac-address=80:2A:A8:C6:AD:EF interface=ether1 bridge=bridge-LAN on-interface=ether1

3 DL  mac-address=C4:AD:34:05:B5:1E interface=ether1 bridge=bridge-LAN on-interface=ether1

4 DL  mac-address=C4:AD:34:05:B5:1F interface=ether2 bridge=bridge-LAN on-interface=ether2

5 DL  mac-address=C4:AD:34:05:B5:20 interface=bridge-LAN bridge=bridge-LAN on-interface=bridge-LAN
[admin@MikroTik] >
```

Activate Windows
Go to Settings to activate Windows.

- Investigated repeated inform/decryption messages and controller adoption visibility.

```
>> netsh interface portproxy show all

Status   Name              DisplayName
------   ----              -----------
Running  iphlpsvc          IP Helper

Listen on ipv4:            Connect to ipv4:

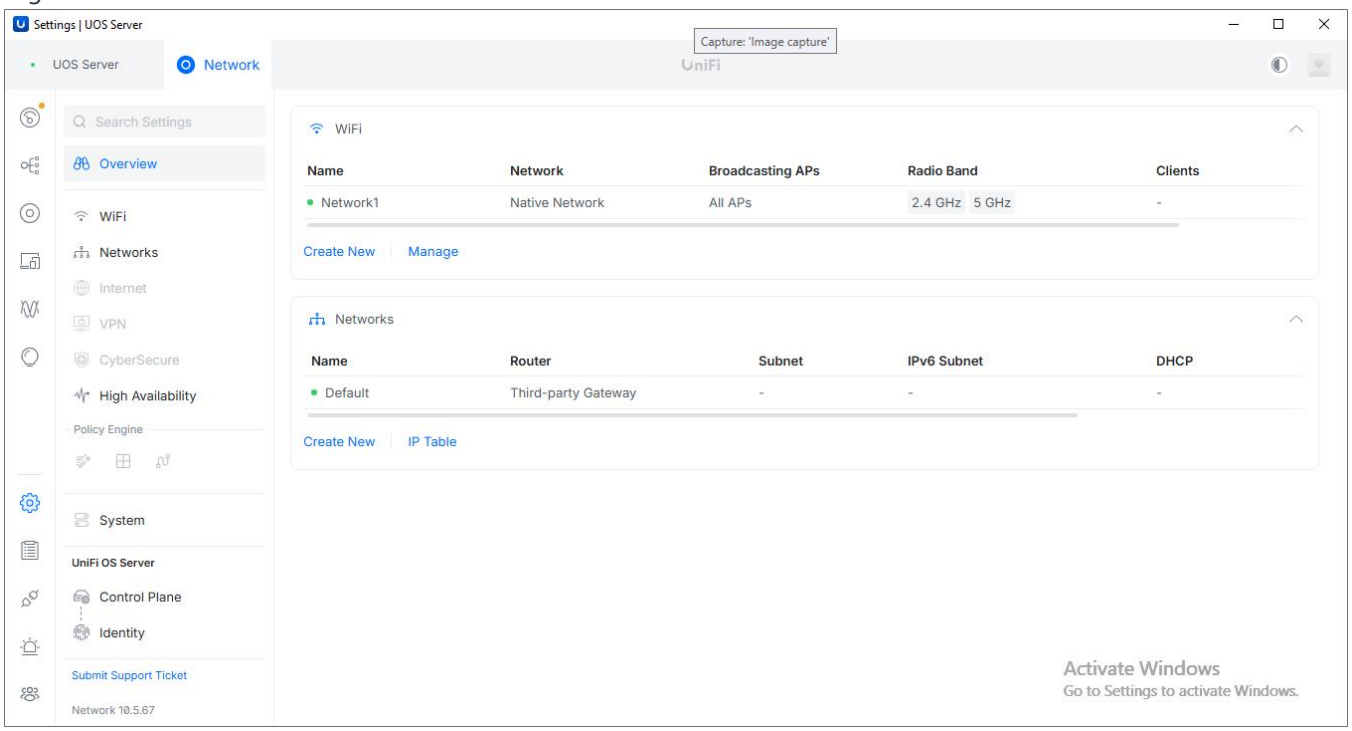
Address         Port       Address         Port
--------------- ---------- --------------- ----------
0.0.0.0         10003      192.168.47.48   10003
0.0.0.0         3478       192.168.47.48   3478
0.0.0.0         5514       192.168.47.48   5514
0.0.0.0         5671       192.168.47.48   5671
0.0.0.0         6789       192.168.47.48   6789
0.0.0.0         8080       192.168.47.48   8080
0.0.0.0         8444       192.168.47.48   8444
0.0.0.0         8881       192.168.47.48   8881
0.0.0.0         8882       192.168.47.48   8882
0.0.0.0         8880       192.168.47.48   8880
0.0.0.0         28082      192.168.47.48   28082
0.0.0.0         9543       192.168.47.48   9543
0.0.0.0         11443      192.168.47.48   11443
```

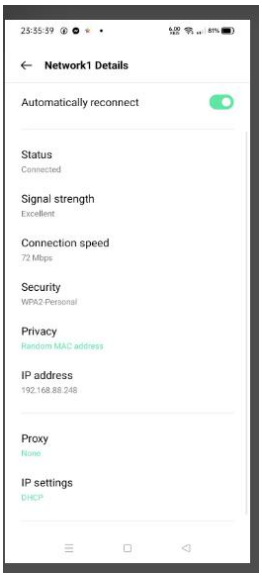
- Investigated a Windows port conflict affecting UniFi Network Server's inform port; verified the service later bound the required ports.

7. Final Result / Current Status

AP LAN reachability and Wi-Fi broadcasting were verified during the lab. Controller adoption and VLAN-backed main/guest SSID

segmentation .





8. What I Learned

- Difference between AP IP reachability, controller adoption, and successful client traffic.
- How to use ping, ARP, SSH, device information, and controller logs together.
- SSID-to-VLAN segmentation needs consistent settings across the AP and upstream network.

9. Real-World Application

Relevant to small-office Wi-Fi deployment, AP onboarding, controller troubleshooting, and guest/main wireless separation.

10. Evidence Checklist

☑ Topology diagram with labels and IP/VLAN plan

- ☑ Configuration screenshot or sanitized command output
- ☑ Verification output showing successful tests
- ☑ Troubleshooting evidence: symptom → investigation → fix → retest
- ☑ Date, device/software version, and brief caption

For more information about my troubleshooting experience , you may see it to my files thank you!